How far have deep learning models come on visual reasoning?

Abstract

Large-scale pretraining has become the dominant paradigm in computer vision, but do the representations it produces support visual reasoning—the ability to trace contours, group elements, and resolve spatial relationships? These capabilities have proven remarkably difficult to learn: neither CNNs nor transformers could solve benchmarks like Pathfinder when trained directly on millions of examples (Linsley et al., 2018; Kim et al., 2020; Tay et al., 2021). Here, we ask whether large-scale pretraining on natural images succeeds where supervised training on these tasks does not. We probe 1,259 pretrained models from the TIMM library on Pathfinder and cluttered ABC (cABC), using linear readouts from frozen representations. Most models fail as the required reasoning range increases. However, the largest and most recent Vision Transformers break this trend, showing a strengthening relationship between ImageNet accuracy and visual reasoning performance. These results suggest that large-scale pretraining may give rise to long-range reasoning capabilities that explicit training could not—with implications for architecture design and the use of deep networks as models of biological vision.

Introduction

Visual reasoning—tracing contours, grouping elements, resolving spatial relationships—requires integrating information across distant locations in the visual field. Theories of visual perception draw a distinction between parallel, feedforward grouping of local features and serial, attention-demanding grouping of spatially distant elements (Houtkamp & Roelfsema, 2010). Most vision models are optimized for tasks like object recognition and scene classification, where local features carry the discriminative signal. Whether these models also acquire long-range reasoning capabilities, particularly through large-scale pretraining, is an open question.

Prior work suggests they should not. Benchmarks designed to isolate long-range reasoning—Pathfinder (Linsley et al., 2018; Kim et al., 2020) and cluttered ABC (cABC)—parametrically control the spatial range over which reasoning must occur. Neither CNNs nor transformers could solve these tasks when trained directly, even with millions of examples; the Long-Range Arena confirmed that no transformer variant learned Pathfinder reliably (Tay et al., 2021). Since then, the architectural landscape has expanded: Vision Transformers (Dosovitskiy et al., 2021) introduced global self-attention over image patches, and state-space models (Gu et al., 2022) extended recurrent architectures to long sequences. Both offer, in principle, the capacity for long-range spatial interaction that CNNs lack. But it remains unclear whether these advances translate into genuine reasoning ability, or simply provide better features and more capacity.

We test this by probing 1,259 pretrained models from the TIMM library (Wightman et al., 2021) on Pathfinder and cABC at multiple difficulty levels, using linear readouts from frozen representations.

Methods

Models. We sourced 1,259 pretrained models from the PyTorch Image Models (TIMM) library (Wightman et al., 2021), spanning four architecture families: convolutional neural networks (CNNs), Vision Transformers (ViTs), convolutional-transformer hybrids (ConvViTs), and recurrent neural networks (RNNs). All models were pretrained for image classification and their weights were frozen during evaluation.

Tasks. We evaluated models on two visual reasoning challenges designed to parametrically control the spatial range over which reasoning must occur. In Pathfinder, participants judge whether two white markers lie on the same or different curve; difficulty increases with curve length. In the cluttered ABC (cABC) task, participants judge whether two markers fall on the same or different letter; difficulty increases with the independence in alignment and orientation between letters. For both tasks, we trained and evaluated linear probes at each difficulty level independently, allowing us to track how performance is affected by the required spatial range of reasoning.

Evaluation. For each model, we extracted frozen penultimate-layer representations and trained a linear probe to classify stimuli at each difficulty level. Linear probing was chosen for two reasons. First, it allowed us to scale evaluation across the full set of 1,259 models. Second, training from scratch has been shown to overestimate architectural differences (Amos et al., 2024); probing pretrained representations instead directly tests whether features learned for image classification generalize to long-range visual reasoning.

Results

Across both tasks, aggregate model performance declined as difficulty increased, confirming that the spatial range of reasoning is a key bottleneck for pretrained vision models (Fig. 1a). However, the severity and pattern of this decline differed markedly between tasks.

On Pathfinder, models achieved above-chance accuracy at the lowest difficulty, but performance collapsed to near-random levels at the highest difficulty across nearly all 1,259 models. Critically, the few models that achieved above-chance performance at higher difficulties were exclusively those with the highest ImageNet Top-1 accuracy—predominantly large-scale ViTs. CNNs were uniformly weak across all Pathfinder difficulty levels. Transformer-based architectures fared better, though performance still degraded as the required range grew.

On cABC, models performed significantly better overall, and the decline with difficulty was more gradual. A broader range of architectures, including CNNs, achieved above-chance performance even at higher difficulty levels. Nevertheless, CNNs exhibited the steepest drop-off as difficulty increased, mirroring the architectural pattern observed in Pathfinder.

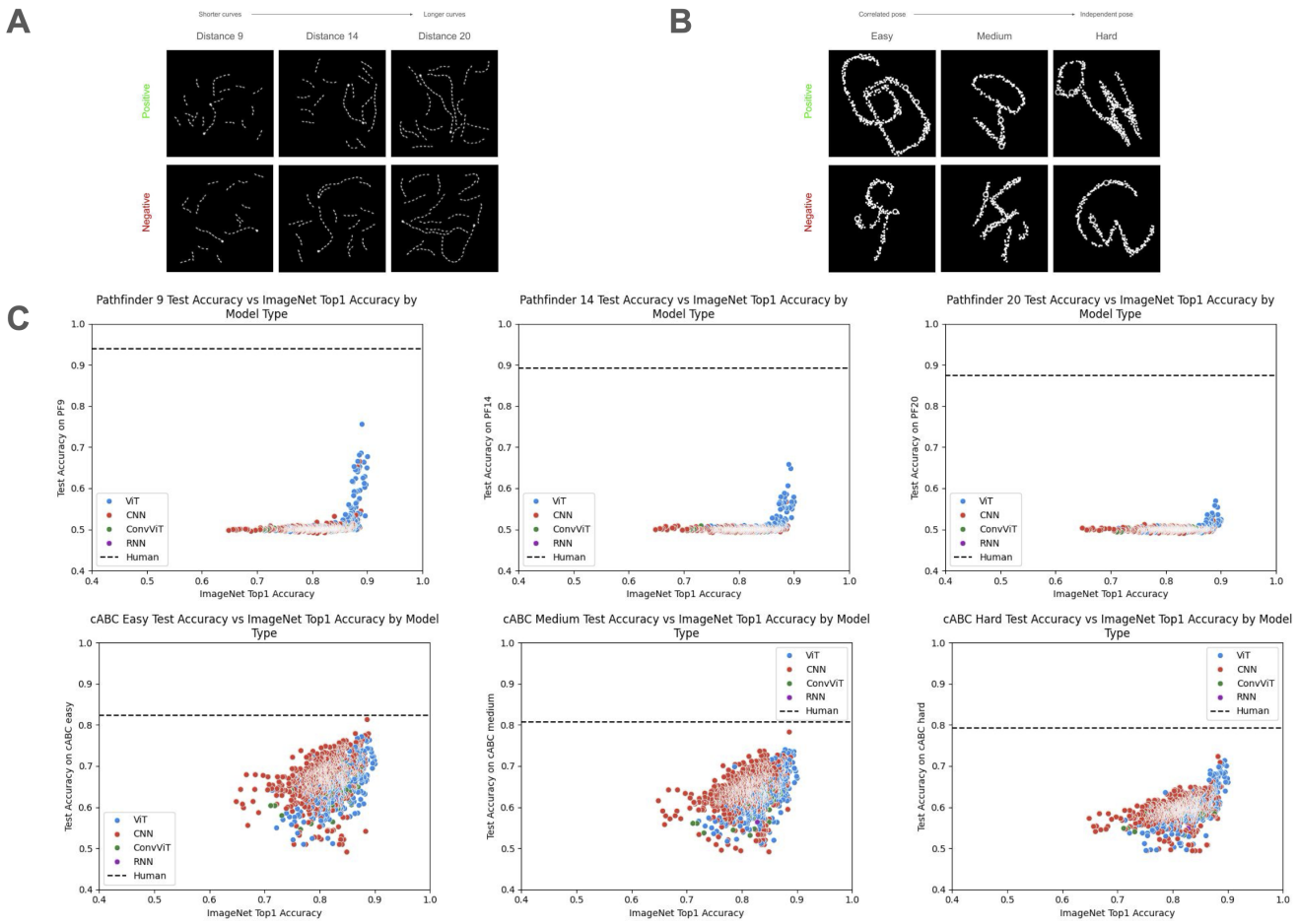

Figure 1: **Model performance on PathFinder and cABC challenges. A.** Examples of PathFinder, where participants must determine whether two white markers fall on the same or different path. **B.** Examples of cABC, where participants must determine whether two white markers fall on the same or different letter. **C.** On both challenges, model performance (p = 1259) noticeably decreased as the difficulty of the task—a control for the distance in the long-range reasoning required for the task—increased, indicating that models struggle to generalize to longer-range reasoning problems after being trained on natural images.

Conclusion

Most pretrained vision models fail to encode long-range spatial relationships, even when they achieve high accuracy on standard classification benchmarks. The exceptions are the largest and most recent Vision Transformers, which show a strengthening relationship between ImageNet accuracy and visual reasoning performance. This suggests that sufficient scale and modern pretraining may give rise to long-range reasoning capabilities without explicit training.

From a neuroscience perspective, these findings speak to a longstanding distinction between feedforward and recurrent processing in biological vision (Houtkamp & Roelfsema, 2010). Tasks like Pathfinder are thought to require iterative, potentially recurrent computation in the visual cortex. The fact that only the largest transformers—with global self-attention over the full image—succeed where CNNs uniformly fail is consistent with this view: architectural capacity for long-range interaction appears necessary, and local feature extraction is not sufficient. Whether these models solve such tasks through mechanisms that parallel biological recurrence or through qualitatively different strategies remains an open question for future work.

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